Project Imaging-X: A Survey of 1000+ Open-Access Medical Imaging Datasets for Foundation Model Development

Zhongying Deng, Cheng Tang, Ziyan Huang, Jiashi Lin, Ying Chen, Junzhi Ning et al. | General Medical AI

https://arxiv.org/abs/2603.27460

Research Questions

- What is the current landscape of publicly available medical imaging datasets and how are they distributed across modalities, tasks, and anatomical regions?
- How can fragmented medical imaging datasets be systematically integrated to create larger, more coherent resources for foundation model training?
- What are the key gaps and limitations in existing medical imaging datasets that hinder the development of versatile medical foundation models?

Core Idea

Medical foundation models lag behind general AI models primarily due to data scarcity - while general models train on billions of images, medical models use only millions. The core problem is extreme fragmentation: thousands of small, isolated datasets scattered across narrow tasks and modalities. The key insight is that this fragmentation can be turned into scale through systematic integration. The authors propose a metadata-driven fusion paradigm (MDFP) that identifies datasets with shared characteristics (modality, anatomy, task) and combines them into larger, more balanced training resources, effectively transforming many small data silos into fewer but much larger unified datasets.

Key Figure

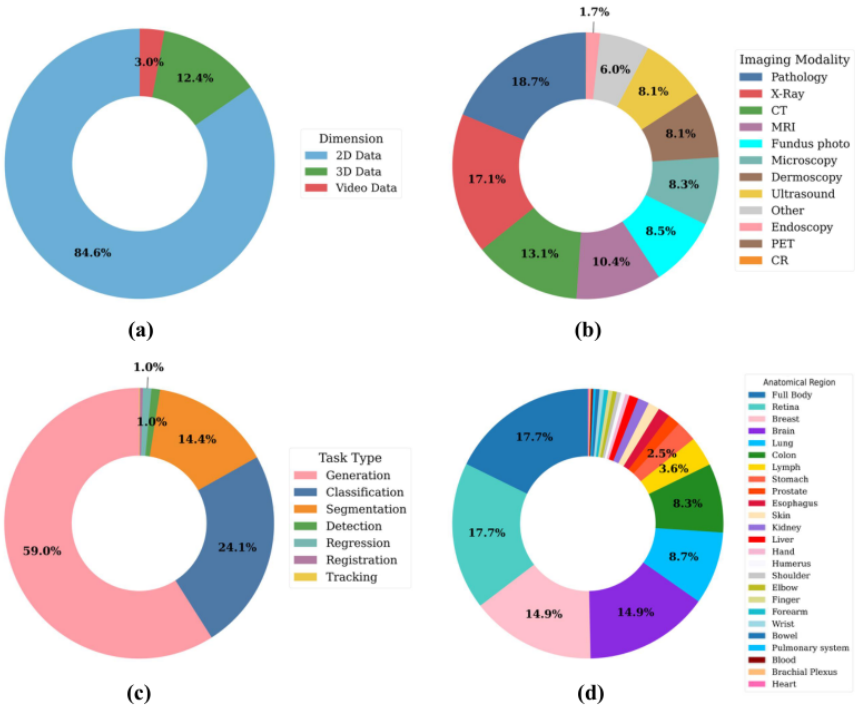

Figure 7: The distribution of (a) imaging dimensionalities, (b) modalities, (c) tasks, and (d) anatomical regions.

Figure 7: Shows the stark imbalance in medical imaging datasets - 84.6% are 2D images, with heavy skew toward pathology (18.7%) and X-ray (17.1%) modalities, and classification (39.0%) dominating tasks, revealing critical gaps for foundation model development.

Limitations

- MDFP relies on metadata quality and availability, which may be inconsistent across datasets
- Integration process does not address fundamental issues like annotation quality harmonization or domain shift between datasets
- Survey limited to open-access datasets, potentially missing large proprietary collections
- No validation of whether integrated datasets actually improve foundation model performance compared to individual datasets